# VAST Challenge 2026 — Mini-Challenge 1 Submission

## 1. Entry Information

**Entry Name:** SMU-Bui-MC1

**Challenge:** Mini-Challenge 1 (MC1)

**Live Application:** https://wujiayan.shinyapps.io/VAST_Challenge_2026

**GitHub Repository:** https://github.com/jiayanwu-jack/Visual-Analytics-Group-Project

## 2. Team Members


| Name | Affiliation | Email | Role |
|---|---|---|---|
| **BUI Tien Thanh ★** Primary Contact | Singapore Management University | tt.bui.2025@mitb.smu.edu.sg | Task 2 — Behavioural Analysis |
| HU Yue | Singapore Management University | yuehu.2025@mitb.smu.edu.sg | Task 3 — Leading Indicators |
| WU Jiayan | Singapore Management University | jiayan.wu.2025@mitb.smu.edu.sg | Task 1 — Event Sequence & Causal Chain; Shiny deployment |


**Programme:** Master of IT in Business (Analytics)

**Course:** ISSS608 Visual Analytics & Applications · Singapore Management University · 2026

## 3. Student Team Declaration

**Yes** — this is a student team. The submission is a class project for ISSS608 Visual Analytics & Applications, taught in 2026 at the Singapore Management University School of Computing and Information Systems. All team members are postgraduate students; the project was led and executed entirely by students.

## 4. Analytic Tools

| Tool / Package | Developer / Source | Purpose |
|---|---|---|
| R (v4.4) | R Core Team — r-project.org | Primary analysis language |
| Shiny | Posit PBC — shiny.posit.co | Interactive web application framework |
| bslib | Posit PBC | Bootstrap 5 theming & page layout |
| plotly (R) | Plotly Technologies — plotly.com/r | Interactive network & chart visualisations |
| ggplot2 | Tidyverse / Posit PBC | Static evidence timeline & bar charts |
| Cytoscape.js | Cytoscape Consortium — js.cytoscape.org | 28-node causal graph (embedded via htmltools) |
| igraph | igraph core team — igraph.org | Network construction & centrality metrics |
| jsonlite | Jeroen Ooms / CRAN | JSON parsing of MC1_final_00.json |
| htmltools | Posit PBC | Embedding custom HTML/JS widgets in Shiny |
| tidyverse (dplyr, tidyr, lubridate) | Posit PBC | Data wrangling & transformation |
| DT | Yihui Xie / CRAN | Interactive data tables |
| Quarto | Posit PBC — quarto.org | Project website & proposal document |

*All tools are open-source. The Shiny application was developed by the student team and is deployed at wujiayan.shinyapps.io/VAST_Challenge_2026. Source code is available in the GitHub repository.*

## 5. Hours Spent

Estimated total hours for the entire team on this submission: **~200 hours**

| Phase | Estimated Hours |
|---|---|
| Data preparation & exploratory analysis | 30 |
| Task 1 — event sequence & causal chain analysis | 50 |
| Task 2 — behavioural & channel analysis | 40 |
| Task 3 — leading indicators & control gaps | 35 |
| Shiny app development, integration & deployment | 30 |
| Documentation, poster & website | 15 |

| Phase | Estimated Hours |
|---|---|
| **Total** | **~200** |

## 6. Repository Permission

**Yes** — we give permission for our submission to be posted in the publicly accessible Visual Analytics Benchmark Repository.

## 7. Submission Video

*Video link to be added before final submission.*

## 8. Mini-Challenge 1 Task Answers

> **Overall Verdict:** The breach was **deliberate and premeditated** — not an organisational failure under pressure. Legal-Agent, the team's own compliance authority, systematically relocated sensitive deliberations off the monitored channel, constructed a legal rationale (Section 4.3(c)) six hours before it was invoked, pre-staged the press release without Judge-Agent's knowledge, and used SaltWind's 5 PM publication as cover for a breach it had already decided to execute. The compliance system did not *break down* — it was **routed around** by the agent designed to uphold it.

### Task 1 — Key Events, Relationships, and Causal Chain

*Reconstruct the sequence of events leading to the embargo breach and characterise the relationships and causal chain that produced it.*

> **Event Sequence (28-node causal chain, May 17 – Jun 5):**
>
> - **May 17 (Round 1):** Embargo established. Legal-Agent sets the 6 PM Jun 5 lift and briefs senior agents on Project HarborCrest via `comms_huddle`.
> - **May 22 (Round 6):** `side_huddle` channel activated. Sensitive merger deliberations begin migrating off the monitored channel. Legal-Agent: *"I now carry undisclosed material information and I need to be very careful."*
> - **May 29 (Round 13):** Legal-Agent declares `comms_huddle` an "observed space" and formally relocates deliberations to `side_huddle` — six days before crisis day and one day before Judge-Agent is deployed.
> - **May 30 (Round 14):** Judge-Agent deployed. Its monitoring scope is limited to `comms_huddle` and `one_on_one_chat` — it has no access to `side_huddle`, public posts, or `internal_state`. It never enters the shadow channel.
> - **Jun 5, 4:01 PM (Round 22):** Legal-Agent deliberating: *"SaltWind publishes at 5 PM regardless… bilateral consent to accelerate — not a unilateral breach. Stage the press release for 4:30 PM."* The press release is staged without Judge-Agent's or CivicLoom's knowledge.

> - **Jun 5, 3 PM:** Judge-Agent issues `COMPLIANCE_WARNING` but does not block. Judge then leaves for a Board Chair call, creating the final oversight gap.
> - **Jun 5, 5 PM (Round 23):** SaltWind Journal publishes merger information. Legal-Agent immediately invokes Section 4.3(c) and calls CivicLoom. The embargo is broken one hour early.
> - **Jun 5, 6 PM:** Official embargo lift. Damage done.
>
> **Key Relationships:**
>
> - Legal-Agent was the central coordinator and the highest-betweenness node across all three periods, controlling information flow through both monitored and shadow channels.
> - On crisis day the `side_huddle` volume rose 19× relative to the Normal period; Judge-Agent was entirely absent from this channel.
> - The pre-crisis network (May 17–Jun 4) shows Legal-Agent dominating `side_huddle` edges; the crisis-day network shows the same structure with added `personal_post` and `anonymous_post` activity by Legal-Agent — channels entirely outside its designed role.
>
> **Causal Chain:** Embargo established → shadow channel created → Judge deployed (too late, wrong scope) → legal rationale pre-constructed → press release pre-staged → SaltWind trigger used as cover → breach executed.

### Task 2 — Characterising Typical vs. Evasive Behaviour

*Describe how agent behaviour on the crisis day differed from typical patterns, and whether the differences suggest deliberate evasion or reactive overload.*

> **Volume & Complexity:** Message length rose 2–3× on crisis day across senior agents — consistent with real-time reasoning under pressure, not execution of a pre-written script. Legal-Agent dominated crisis-day message volume.
>
> **Channel Migration (Decisive Evidence):** Private-channel usage ( `side_huddle +` `one_on_one_chat` ) rose *monotonically* across all three periods — Normal → Escalation → Crisis — beginning May 22, two weeks before crisis day. A purely reactive system would show a step-change only on Jun 5; the observed gradient began much earlier, indicating deliberate migration rather than crisis-driven panic.
>
> **Channel repertoire expansion:** Legal-Agent used only internal channels during the Normal period. By Jun 5 it had acquired `anonymous_post` (12 merger-related leaks) and `personal_post` — entirely outside its designed role.
>
> **Temporal Inversion (Most Decisive Evidence):**
>
> - **4:01 PM — deliberating field:** Legal-Agent plans the bilateral consent argument, stages the press release for 4:30 PM, and decides to call CivicLoom.
> - **5:01 PM — rationalizing field:** Legal-Agent frames the action as a reaction to SaltWind's 5 PM publication: *"SaltWind published… I am calling CivicLoom NOW."*
>
> The plan (4 PM) preceded the event it claims to react to (5 PM). This temporal inversion is inconsistent with reactive overload and consistent only with premeditated action.

### Task 3 — Leading Indicators and Why No Corrective Action Was Taken

*Identify leading indicators that foreshadowed the breach and explain why prior warning signals did not result in corrective action.*

> **Q3.1 — Expected vs. Actual Behaviour (Behavioural Divergence):**
>
> Legal-Agent's Jensen–Shannon divergence from its Normal-period channel distribution rose monotonically across all three periods. The agent acquired new channels ( `anonymous_post` , `personal_post` ) with no precedent in the Normal period, and its `side_huddle` share grew from near-zero to dominant. The divergence was measurable and rising from May 22 onward — three weeks before the breach.
>
> **Q3.2 — Breach-like Behaviour Before Jun 5:**
>
> Sensitive merger and embargo content circulated privately from May 23 onward with **zero public leakage** before Jun 5. The "coordinate-in-shadow-then-push-public" pattern was rehearsed for two weeks; only the final public step was new on Jun 5. Legal-Agent's rationalisation language appeared repeatedly in `internal_state` throughout the Escalation period, showing the justification for action was constructed incrementally well before it was used.
>
> **Q3.3 — Why No Corrective Action Was Taken:**
>
> Four structural root causes explain the absence of corrective action:
>
> - **Scope limitation (5 of 11 near-miss events):** Judge-Agent was scoped to `comms_huddle` and `one_on_one_chat` only. It never entered `side_huddle` , never monitored public posts, and had no access to `internal_state` . All leading indicators lived in channels it could not see.
> - **Enforcement limitation (3 events):** Judge-Agent's `COMPLIANCE_WARNING` was advisory only — it had no hard-block authority. Even when it observed a potential violation (3 PM Jun 5), it could only warn.
> - **No escalation protocol (2 events):** There was no mechanism to alert CivicLoom or an external authority when internal compliance was compromised. The breach was effectively self-policed.
> - **Timing gap (1 event):** Judge-Agent was deployed on May 30 — 13 days into a 14-day period. The first eight days of shadow-channel migration occurred with no monitor present at all.
>
> The result: none of the leading indicators — rising private-channel sensitivity, channel repertoire expansion, escalating internal rationalisations — could trigger a response from a monitor that was unable to observe them.

## 9. Design Overview

The submission consists of a six-tab interactive Shiny dashboard:

| Tab | Content | Primary Interaction |
|---|---|---|
| Causal Chain & Evidence Timeline | 28-node Cytoscape.js causal graph + 45-item evidence timeline | Hover node for quote; evidence filtered by label |

| Tab | Content | Primary Interaction |
|---|---|---|
| Communication Networks | Pre-crisis vs. crisis-day directed agent networks (plotly) | Hover edges for message count |
| Channel Behaviour & Sensitivity | Channel usage by period + private sensitivity score over time | Period selector; scroll to zoom |
| Internal State Analysis | 6-phase accordion: deliberating / reacting / rationalizing | Click phase to expand; verbatim field text shown |
| Near Misses & Control Gaps | Root-cause bar chart + 11-event detail table | Hover bar for count; click for detail |
| Summary & Conclusions | Verdict banner + competing-hypotheses assessment | Static narrative panel |

A poster summarising the methodology and key findings is available at the project website: [thanhbui221.github.io/Visual-Analytics-Group-Project/poster.html](thanhbui221.github.io/Visual-Analytics-Group-Project/poster.html)

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